

PART I

The Circle Is the Plumbing, Not the Fraud

Bloomberg's 27 July map of circular AI deals is a fair picture of a real risk and a misleading picture of its cause. The reciprocity exists because capital is the only abundant input in the AI buildout — and because the entities with the demand have no credit rating. What determines the outcome for owners is not the size of the arrows. It is the instrument, the terms, and who is contractually unable to stop.

DATA AS AT THE 28 JULY 2026 CLOSE • BALANCED ASSESSMENT; VIEW STATED IN §6 • PREPARED FOR INTERNAL DISCUSSION

BOTTOM LINE

- **The circularity is real and it is disclosed.** Nvidia's equity book in its own customers went from \$3.4bn to roughly \$40bn in one fiscal year; FY26 other income was \$11.1bn against \$2.6bn, i.e. ~8% of pre-tax income from marking up its ecosystem. None of this is hidden. That is the material difference from 1999–2002, when the round-trips were concealed and reversed through revenue.
- **It is also, in large part, structurally unavoidable.** The binding constraints in 2026 are memory, advanced packaging, substrate film, turbines, transformers and interconnect queues — not money. Nvidia earns \$130bn of operating income on capex of 2.8% of revenue; its customers need to commit capital three to five years ahead of revenue with no investment-grade rating. The vendor balance sheet is the bridge.
- **The instrument is migrating from equity to credit, and that is the event that matters.** Nvidia's 10-K states plainly that it has entered into no financing arrangements; its entire disclosed guarantee book is \$3.5bn. The reported ~\$250bn Ohio backstop would be ~71× that, and would move it from the GE position (owner) into the Lucent position (lender). Its 5-year CDS hit a record 82bp on 27 July. The credit market repriced before the equity did.
- **History says overbuilds of this type are productive for the economy and value-destroying for the funders** — railways, electrification, radio and fibre all resolved that way. The discriminating variable was never demand or scale. It was capital structure, and specifically the presence of non-cancellable forward commitments. There are roughly \$662bn of hyperscaler leases signed but not yet commenced, on no balance sheet.
- **Our read: the nearest analogue is 4G, not Lucent and not Enron** — real continuous consumption, adequate asset life, majority-internal funding at the top of the stack, and a live, unresolved question about whether the party writing the cheques is the party that gets paid. We would underwrite the compute demand and refuse to underwrite the funders indiscriminately. Monitorables in §5.

1 WHAT THE CHART ACTUALLY SHOWS

The chart is reproduced as Exhibit 1 at the end of this note. Strip out the density and there are two flows that matter: hardware moving out from the suppliers, and money moving in to the buyers — frequently between the same two parties. Nvidia sits at the centre because it does both simultaneously. It sells accelerators to OpenAI, Anthropic, xAI, CoreWeave, Nebius, Nscale and Mistral, and it holds equity in most of them: up to \$10bn into Anthropic, \$30bn into OpenAI's March round, \$5bn into Intel, ~\$2bn into the xAI Colossus 2 GPU vehicle, a further \$2bn into CoreWeave in January, a 9.3% passive stake in Nebius filed on 22 July, and a reported investment into Safe Superintelligence announced on 27 July. Its non-marketable equity securities went from \$3.4bn to \$22.3bn across FY2026 and to \$42.3bn by the April quarter, with roughly \$18–30bn more in listed holdings.

The same pattern runs across the rest of the map. Google agreed on 24 April to invest up to \$40bn in Anthropic in cash and compute alongside up to 5GW of dedicated TPU capacity; Anthropic then rents that capacity. Microsoft and Nvidia put up to \$5bn and \$10bn respectively into Anthropic in November, against Anthropic's \$30bn Azure commitment. Amazon put ~\$50bn into OpenAI's February round; OpenAI models appeared on Bedrock a day after the Microsoft exclusivity was removed. AMD issued warrants for 160m shares at \$0.01 to OpenAI and a second identical tranche to Meta — some 320m shares, close to 20% of the company, handed to two customers. Broadcom wrote residual-value guarantees covering \$30bn of the Anthropic TPU vehicle. Google itself has written ~\$44bn of data-centre lease guarantees in nine months on buildings it will never own, plus warrants for up to 14% of TeraWulf.

Two corrections to the chart, both minor but worth noting for anyone quoting it. Nvidia is no longer a \$5trn company or the most valuable one: it closed 28 July at \$197.01, a market capitalisation of ~\$4.77trn, having ceded

first place to Apple on 27 July. And SpaceX at ~\$1.5trn is a public market capitalisation post its \$75bn June IPO, not a private mark. The chart's publication date is itself the story: 27 July was the day Nvidia fell 4.99% on the Ohio backstop reports.

2 WHY INVESTORS ARE UNEASY – FOUR SEPARATE MECHANISMS

i. Revenue you funded yourself is not independent evidence of demand

The concern is not that the transactions are fake. It is that the marginal buyer's ability to pay was partly created by the seller, so the order book is not a clean read on end demand. Nvidia's Q1 FY2027 GAAP net income of \$58.3bn exceeded its operating income of \$53.5bn, because \$13.4bn of unrealised gains on listed stakes flowed through below the line. Alphabet's Q2 print booked \$98.0bn of net other income from unrealised marks on Anthropic and SpaceX, taking EPS up 294%. Nvidia's cumulative gross unrealised gains on non-marketable securities stand at \$2.7bn against cumulative impairments of \$176m — a 15:1 asymmetry that cannot survive a draw-down. Its own 10-Q discloses that "one AI research and deployment company contributed to a meaningful amount of our revenue by purchasing cloud services from our customers."

ii. Concentration dressed as diversification

Fifteen nodes on the chart, exposed to the same one or two counterparties.

TABLE 1 – COUNTERPARTY CONCENTRATION, AS FILED

DISCLOSURE	FIGURE	SOURCE / PERIOD
Nvidia – two direct customers, share of revenue	36%	FY2026 10-K (yr to 25 Jan 26)
Nvidia – three direct customers, share of receivables	64%	Q1 FY2027 10-Q (up from 56% q/q)
Oracle – remaining performance obligations	\$638bn	FY2026, +363% y/y
– of which OpenAI, per S&P	~50%	S&P downgrade rationale, 9 Jul 26
Oracle market capitalisation, for scale	\$345bn	28 Jul 26 close; –65% off high
Oracle FY2026 free cash flow	–\$23.7bn	On capex of \$55.7bn
OpenAI 2025 revenue / operating loss	\$13.1bn / –\$20.9bn	Audited, reported Jun 26

Oracle was cut to BBB–, one notch above sub-investment grade, on 9 July 2026; its 5-year CDS is at an ~18-year high. Its backlog now exceeds its own equity value, and roughly half of it sits with a single unrated counterparty.

iii. The instrument is migrating from equity to credit

This is the genuinely new risk and it is what the market repriced in late July. An equity stake caps the loss at carrying value, lands in non-operating income, and never inflated revenue in the first place. A guarantee makes you a subordinate, unsecured, concentrated lender to the exact demand forecast you are selling into. That is the distinction between GE behind the utilities in the 1920s — which survived as an owner — and Lucent, Nortel, Cisco and Motorola behind the CLECs in 2000, which took the losses as lenders (Lucent provisioned ~\$2.8bn against customer financing and wrote off ~\$2.6bn; Cisco carried a 58% reserve against gross lease receivables). Nortel's own FY2001 10-K explains the trap: the financing was originated on the assumption it could be syndicated to banks, and when the credit window shut the vendor became involuntary lender of last resort to its own order book. Peak exposure was \$5.2bn, 79% of it undrawn, and it still produced a \$19.4bn quarterly loss.

THE SINGLE MOST IMPORTANT BINARY IN THE FILE

Nvidia's FY2026 10-K, filed for the year to 25 January 2026, states: "*We have entered ... long-term capacity purchase obligations and financial guarantees, and have been asked to offer financing arrangements to support our customers' ... buildout. **We have not entered into any financing arrangements.***" Its Q1 FY2027 10-Q caps maximum gross exposure under all existing facility-lease guarantees at **\$3.5bn**. The reported Ohio structure — a ~\$250bn guarantee of lease and construction debt for a 10GW OpenAI campus developed by SB Energy, with parallel talks on financing up to \$350bn of chip purchases — would be roughly **71×** the disclosed guarantee book, would carry no equity upside, and is being contemplated precisely *because OpenAI has no investment-grade rating*. Talks are early and may collapse. If they do not, the sentence above stops being true, and the analogue changes from GE to Lucent.

iv. Accounting lag and the off-balance-sheet layer

Moody's found in February that the five largest hyperscalers doubled lease commitments to \$969bn during 2025, of which \$662bn had not yet commenced and therefore sat on no balance sheet — equal to 113% of the same five firms' adjusted debt. Trailing four-quarter capex through March 2026 was \$433.9bn against roughly \$149bn

of recognised depreciation. A single year's cohort adds \$65–75bn of annual depreciation at six-year lives and \$95–110bn at four. Reported margins compress mechanically into 2027–29 whatever AI revenue does. Amazon's capex is now 102% of operating cash flow and its trailing free cash flow has fallen from ~\$26bn to ~\$1.2bn; Alphabet posted –\$5.9bn of quarterly free cash flow. FASB's advisory council has concluded that "urgent standard setting activity by the Board is not needed" while explicitly asking whether disclosure is sufficient — including whether these transactions are arm's length. The disclosure has not been provided.

3 WHY IT NONETHELESS HAS TO BE DONE

Capital is the only abundant input

The bubble framing assumes money is chasing imaginary demand. The 2026 evidence is that money is chasing scarce physical capacity. Interconnection queues run a median of over five years to commercial operation. GE Vernova's gas turbine backlog and slot reservations reached 116GW and it is now taking reservations for 2031; Siemens Energy is booked out to FY2028. Large power transformers average ~128 weeks, with specialised units quoted to four years. Ajinomoto Fine-Techno holds roughly 95% of global ABF substrate film supply, notified a ~30% price increase effective Q3 2026, and its third plant is targeted for 2032. DRAM contract prices rose 90–95% quarter-on-quarter in Q1 2026 and a further 58–63% in Q2; Microsoft attributed ~\$25bn of its ~\$190bn capex purely to component inflation. Northern Virginia data-centre vacancy is 0.3%, an all-time low, with 96% of 2026 deliveries pre-committed. Contrast the fibre analogue: 8% of installed strands were lit in 2001 and still only ~15% by 2006.

The balance sheets are mismatched to the risk, and reciprocity is the bridge

Nvidia generated \$130.4bn of operating income in FY2026 on capex of \$6.0bn — 2.8% of revenue. It is the most asset-light dominant supplier in modern industrial history and has nothing productive to do with the cash internally. Its customers have the inverse problem: triple-digit revenue growth, no credit rating, and a requirement to commit capital three to five years before the revenue lands. No bank will lend twenty-year money to an unrated counterparty against hardware collateral. So either the vendor's balance sheet carries the timing risk, or the capacity is built several years later by someone with a worse view of the demand curve. There is also a real information argument that deserves engaging rather than dismissing: the party best placed to underwrite AI compute demand is the party that sees every order book in the industry. Adverse selection runs backwards from the usual case.

For the labs, compute is cost of goods sold, not capex

Huang's formulation — without compute there are no tokens, and without tokens there is no revenue — is an accounting statement, not a slogan. Under-buying compute is directly foregone revenue in the current period. Anthropic went from roughly \$9bn of run-rate revenue exiting 2025 to \$30bn on Amodei's 8 May statement and a reported \$47bn alongside its May round at \$965bn. Google's token throughput went from 9.7trn per month in mid-2024 to 3.2 quadrillion per month by May 2026. Amodei's own warning that a single year of miscalculated growth could bankrupt the company cuts both ways: for a lab in a winner-take-most race with four to six credible players, the asymmetric error is under-provisioning. Over-committing is individually rational for every participant, and vendor support is what makes that individually rational strategy financeable. The systemic risk and the individual logic are both correct simultaneously.

The shape of demand has changed in a way that favours the assets

Pre-training scaling shows deteriorating cost-elasticity; the active axis is reinforcement learning, post-training and test-time compute, which raise compute per query by an order of magnitude or two and are embarrassingly parallel. They tolerate weak interconnect, geographic distribution and previous-generation silicon. That is why A100s launched in 2020 are sold out in year six with rental rates up ~15% year-to-date, why Google runs seven-to-eight-year-old TPUs at full utilisation, why Azure retirements have historically landed at seven to nine years, and why H100 one-year contract rates went from ~\$1.70/hr in October 2025 to ~\$2.65/hr by March 2026. The nascent GPU forward curve prices total rental decline of only 8–15% over 36 months. Demand is continuous consumption, not one-time provisioning against a capacity forecast — the strongest single structural break from fibre, and the one that most cleanly supports the useful-life assumptions the bears attack.

4 THE HISTORICAL SCORECARD

The uncomfortable finding from every comparable episode is that the asset gets used, the economy wins, and the funders lose. British railways: every mile built was used, revenue rose 2.8× to 1860, and equity fell 64% while net returns came in at 3–4% against a 10% expectation — *The Economist* in 1855 called them "an honour to our age and country: commercially, they are great failures." Radio advertising went from \$4.8m in 1927 to \$215.6m in 1940; RCA shareholders lost 98%. Fibre delivered a 90–99% decline in bandwidth cost and enabled the entire

cloud and streaming economy; equity lost 96–100%, high-yield recoveries were 16 cents, and Cogent acquired \$14bn of assets for \$60m.

The railway detail is worth dwelling on, because it is the case most often cited as a vindication and is in fact the opposite. Mileage in service went from 2,044 in 1843 to 10,433 by 1860 with no abandonment, and yet revenue per mile *fell* 23% between 1845 and 1850 as marginal mileage was pushed into thinner traffic — the precise analogue of falling revenue per fibre-mile, or per gigawatt. Goods revenue only overtook passenger revenue between 1851 and 1853: the industrial, productivity-relevant use case arrived *after* the financial disaster, exactly as internet video did. The asset being used, and used permanently, told you nothing about the return to the people who paid for it.

TABLE 2 – WHAT DISCRIMINATED THE OUTCOMES FOR FUNDERS

CYCLE	FUNDING STRUCTURE	PEAK-TO-TROUGH EQUITY	RECOVERY
UK railways, 1840s	~All equity; charters capped borrowing. Peak-year investment 7.3% of GDP	–64%	From 1850
Utility holding cos, 1920s	Six layers, ~18:1, bonds and guaranteed preferred at every level	–92.7%	34 years
Radio, 1920s	Mostly equity at RCA; value migrated to the franchise	–98%	Decades
Telecom / fibre, 1996–2002	>\$500bn capex, mostly debt; ~\$1trn industry debt stock	–96 to –100%	Never, for the original entities

The Railway Mania is the control experiment. At 7.3% of GDP in 1847 it was roughly five times more intense than peak US telecom investment (~1.6% of GDP in 2000), and it produced a –64% drawdown with no systemic default wave. **Scale did not determine severity. Capital structure did.** Insull's utilities are the mirror image: electric utility revenue fell only 6% in the worst depression on record, and the equity still fell 92.7% and took until 1963 to recover, because fixed charges at every layer absorbed the entire operating margin.

The most transferable insight is subtler than debt-versus-equity, and it is the one we would put in front of anyone modelling the hyperscalers as self-funded. Railway shares were subscribed with a ~5% deposit, with the remaining 95% callable at the directors' discretion and subscribers legally liable for it. Nominal equity funding plus enforceable forward commitments is functionally leverage — which is why UK railway capital formation peaked *two years after* share prices did. Investors could not stop. The 2026 equivalents are the ~\$662bn of leases signed but not commenced, Nvidia's \$95.2bn of supply commitments and \$27bn of cloud commitments, Meta's \$131.1bn of contractual commitments, Amazon's \$84.8bn of unconditional purchase obligations explicitly not reflected on the balance sheet — and the fact that **the cancellability terms of the largest compute contracts are disclosed nowhere.**

TABLE 3 – SCORING 2026 AGAINST THE TESTS THAT MATTERED HISTORICALLY

TEST	VERDICT	EVIDENCE
Funded from the owner's own cash flow	Split	~\$500–600bn internally funded since 2023 (Dallas Fed) is benign; Amazon at 102% of OCF, \$662bn of off-balance-sheet leases, SPVs and private credit are not. Passes for hyperscalers, fails for neoclouds and vehicles.
Paying customers at scale before the build	Fails, less badly than 1999	OpenAI ~\$25bn run-rate on a \$20.9bn 2025 operating loss; CoreWeave \$5.1bn on a \$1.2bn net loss. "Real but unprofitable" beats "revenue-less" — but revenue does not yet cover the commitments.
Asset life vs obsolescence; do old assets cascade?	Leans favourable	A100s sold out at six years; TPUs at 100% utilisation at seven to eight; forward curve –13%/3yrs. Honest counter-precedent is the memory cycle: equipment kept working, the price of its output collapsed.
Pricing power at the point of sale	Currently yes	Rentals up, capacity sold out. But contracted revenue is not pricing power: Equinix has take-or-pay leases and a 3.8–5.3% ROIC, and Digital Realty's operating income has been flat for fifteen years.
Single-owner vs fragmented overbuild	Split by layer	Fails at the model layer — four to six well-capitalised labs buying broadly interchangeable compute. Passes emphatically at the supply layer, which is why Nvidia's returns on capital are a multiple of its customers'.
Disclosure quality as a leading indicator	Fails on the specifics	In every destroyed-capex case the accounting deteriorated <i>before</i> the economics did. Today: cancellability undisclosed; arm's-length character requested by FASB and not provided; impairments 1/15th of unrealised gains; the 2020 S-K amendments deleted the contractual-obligations table.

5 MONITORABLES, IN PRIORITY ORDER

- Does "we have not entered into any financing arrangements" survive Ohio?** Equity stake versus guarantee is the whole distinction between the GE outcome and the Lucent outcome. Watch also whether Nvidia's \$6.3bn CoreWeave unsold-capacity backstop expands, and whether the July revenue-sharing model with Sharon AI and Firmus spreads to weaker counterparties.
- Terms, not size.** The tell that actually predicted 1999 was Nortel financing up to 130% of equipment cost, unsecured, tied to future purchases. Today's version: whether AMD-style penny warrants — 320m shares, ~20% of the company, already issued to two customers — proliferate down the credit curve.

3. **A second hyperscaler shortening useful lives** in a 10-K property-and-equipment estimate-change paragraph. Amazon did it in January 2025 citing "the increased pace of technology development, particularly in artificial intelligence"; nobody followed through the entire FY25 reporting season, and Meta is reportedly extending operational server life from six to seven years. Confirming market signal: H100 neocloud rates breaking below \$2.00/hr.
4. **Backlog conversion and backlog duration.** Oracle at \$638bn of RPO against \$67bn of revenue; CoreWeave at \$99bn against \$12-13bn of guidance. Watch whether backlog grows faster than the revenue behind it, and whether Microsoft's sub-12-month share keeps falling (~40% to ~25% in six months).
5. **First crystallisation events.** The first impairment of a non-marketable stake; the first onerous-contract charge against a compute commitment; the first disclosure of cancellation provisions; the first AMD warrant tranche vesting and whether it is presented as contra-revenue under ASC 606 or as dilution only. The latter is the difference between the headline deal maths and reported revenue.
6. **The credit tape, not the equity tape.** AI-related debt issuance is running at roughly \$570bn for 2026, close to double 2025; private credit to AI has gone from near zero to over \$200bn; nearly 80% of data-centre securities issued since early 2025 quote wider than at issue; order-book cover fell from ~5× in February to under 2× in July, and two bond deals were pulled. Note the contrary signal: CoreWeave's GPU-backed cost of capital compressed ~750bp in 31 months, and its A3 facility abandoned residual-value underwriting for offtaker credit and power availability — lenders are pricing the contract and the electron, not the silicon.
7. **Bottleneck rotation.** ABF film is the least substitutable and least priced-in link. Memory's first flat spot print (1 July) is the signal to watch for a peak. Power constraints are *easing* at the margin — interconnection queue volume fell 10%, the PJM 2028/29 capacity auction cleared 2.5% lower — which means the 2028 ceiling narrative needs re-underwriting rather than repeating.
8. **Investor tolerance for capex**, which visibly cracked on Alphabet's 22 July print: it raised FY26 capex guidance to \$195-205bn and was punished for it. In 2022-23 Amazon cut annual capex from \$63.6bn to \$52.7bn. Discipline exists, and it is the most likely mechanism by which this cycle ends without a default wave.

6 OUR VIEW, AND WHAT WOULD FALSIFY IT

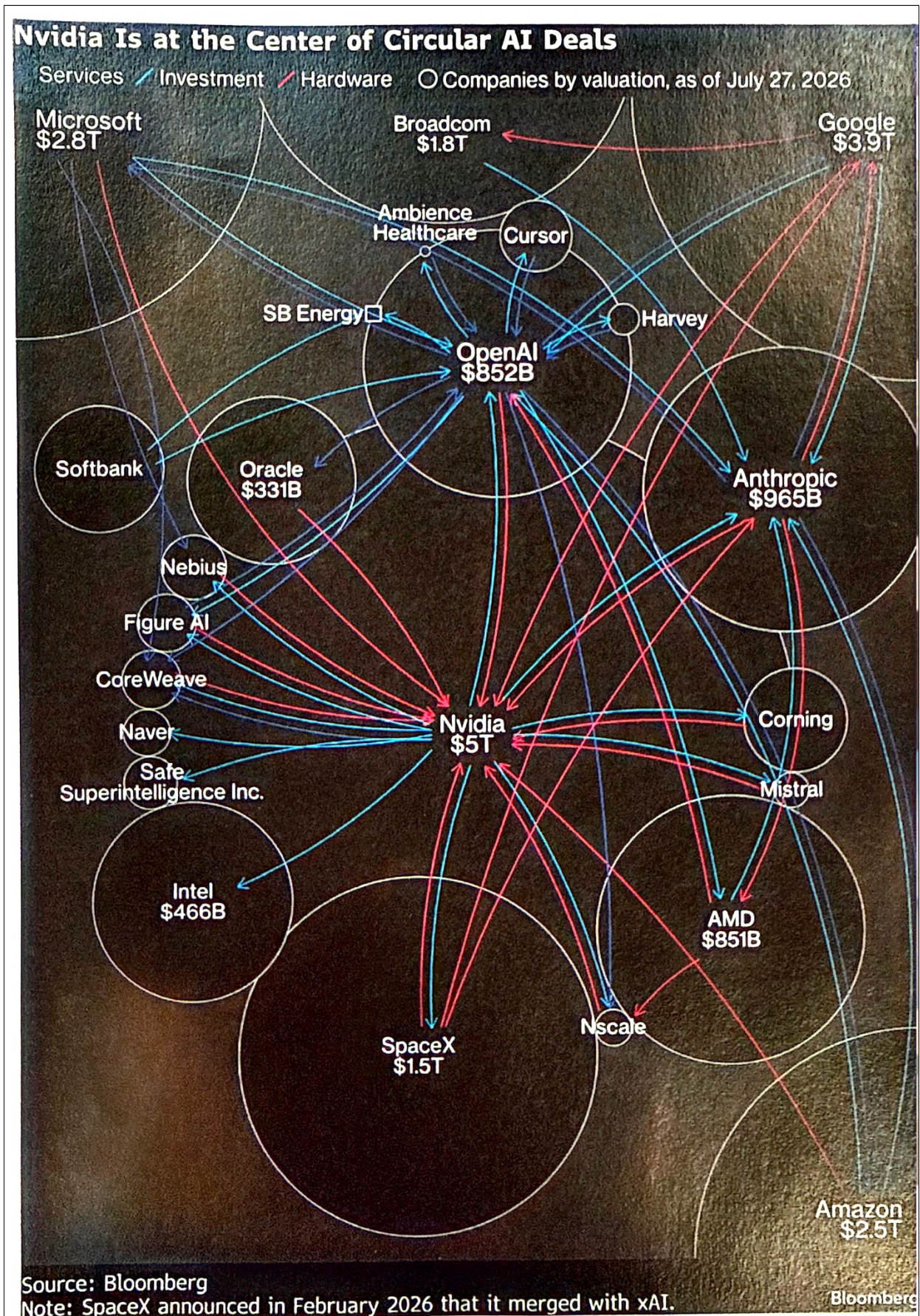
We do not think this is Enron and we do not think it is Lucent. The nearest analogue is 4G: genuine demand, genuine continuous consumption, asset lives that are probably adequate, funding that remains majority-internal at the top of the stack — and a serious unresolved question about whether the party writing the cheques is the party that gets paid. The instrument difference from 1999 is real and material. Equity stakes wipe out through non-operating income rather than reversing through revenue, which makes the eventual damage less violent but also less visible, and considerably slower to show up in the numbers that trigger a repricing.

What follows for positioning is a distinction rather than a direction. We are comfortable underwriting the compute demand: the physical evidence of scarcity is broad, independently sourced and points the same way, and the demand-shape argument in §3 is the strongest structural case against the fibre analogy. We are not comfortable underwriting the funders indiscriminately. In a network overbuild with high fixed cost, near-zero marginal cost and undifferentiated output, competition transfers the productivity gain from asset owners to users, and to a concentrated set of second-generation firms that acquire the assets at post-bankruptcy basis. Google bought its dark fibre in 2005. Somebody will be that buyer this cycle, and it will not be the entity that funded the build.

We would change our mind in either direction on specific, dated, filed evidence rather than on narrative. Toward the bear case: Nvidia converting from owner to guarantor at material scale; a second hyperscaler shortening useful lives; H100 spot rentals below \$2.00/hr; or the first onerous-contract charge on a compute commitment. Toward the bull case: capex discipline arriving voluntarily and being rewarded; the labs demonstrating audited gross-margin-positive inference at scale; and the commitment stack converting to revenue on schedule rather than being renegotiated. The most useful discipline we can impose on ourselves is the one the railway record teaches: being right that an overbuild is under way is not the same as being able to trade it. Lardner published the correct analysis in October 1846 and prices took another three years to bottom, while capital formation kept rising the whole way down, because the commitments were enforceable.

Sources. Primary filings: Nvidia FY2026 10-K and Q1 FY2027 10-Q; AMD FY2025 10-K and OpenAI warrant Ex-4.1; Oracle FY2026 results and the July S&P rating action; Meta and Amazon FY2025 10-Ks; CoreWeave Q1 2026 and DDTL 4.0/5.0 8-Ks; Alphabet Q2 2026. Secondary: Moody's on off-balance-sheet lease commitments, 23 Feb; BIS Annual Economic Report, 28 Jun; Dallas Fed on AI financing, 10 Feb; Morgan Stanley on AI debt issuance, 10 Jun; SemiAnalysis on GPU shortage and phantom data-centre demand; Silicon Data GPU indices and forward curve; LBNL "Queued Up 2026"; PJM 2028/29 capacity auction; CBRE Global Data Center Trends 2026; TrendForce on HBM and DRAM; Bloomberg circular-deals graphic, 27 Jul. Historical: Odlyzko on the Railway Mania; Richmond Fed on telecommunications; Altman & Bana on high-yield defaults; Nortel FY2001 and Lucent FY2002 filings; SEC enforcement releases on Lucent, Qwest, AOL Time Warner and Enron Broadband; Cudahy & Henderson on Insull.

Caveats. The Ohio guarantee is early-stage reporting, not a signed agreement, and may not proceed. Frontier-lab gross margins and burn projections are press-reported or leaked, not audited. Aggregate 2026 hyperscaler capex estimates span \$690-785bn depending on basket; ranges are used where sources conflict. Anthropic's Q2 2026 profitability figure is a company projection, not an audited result. Not investment advice.



Source: Bloomberg, 27 July 2026. Circle area is company valuation. Three edge types: services, investment and hardware. Read it as two flows — hardware moving out from the suppliers and money moving in to the buyers, frequently between the same two parties. Reproduced for internal commentary. Several marks are already stale: Nvidia closed 28 July at ~\$4.77trn, having ceded first place to Apple on 27 July, and SpaceX at ~\$1.5trn is a public market capitalisation following its \$75bn June IPO, not a private mark.